

Comic Craft: Technology Stack Blueprint

The Artist's Toolkit: Core Structural Infrastructure

This document maps the architectural components, execution runtimes, and databases supporting the live production system.

Layer 1: The Pencils (Frontend View Layer)

- **Core Technologies:** React.js, TypeScript, Tailwind CSS, Vite runtime tooling.
- **Role:** Constructing crisp, reactive interface components and running application logic state directly inside client browser viewports.

Layer 2: The Inking & Colors (Backend Service Core)

- **Core Technologies:** Node.js, Fastify Framework, Prisma Object-Relational Mapper.
- **Role:** Authenticating API traffic, evaluating business rules, and driving high-throughput computational updates.

Layer 3: The Vault (Storage & Caching Engines)

- **Core Technologies:** PostgreSQL Database, Redis In-Memory Cache.
- **Role:** Protecting critical transaction tables while scaling read performance with microsecond memory lookups.

Layer 4: The Publishing House (Cloud Infrastructure)

- **Core Technologies:** AWS (ECS Fargate), GitHub Actions, Docker Engine.
- **Role:** Packaging isolated container images and running automated blue-green delivery pipelines.